

Mathematical Logic

- **Exercise 1**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

$\sin\left(\frac{\pi}{4}\right) > 0$ and $\sin\left(\frac{\pi}{4}\right) < 1$,

- **Solution 1.1**

Let:

$p : \sin\left(\frac{\pi}{4}\right) > 0$

$q : \sin\left(\frac{\pi}{4}\right) < 1$

The symbolic logic is:

$$p \wedge q.$$

- **Exercise 2**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

9 is a perfect square or 8 is a perfect square.

- **Solution 2.1**

Let:

$p : 9$ is a perfect square.

$q : 8$ is a perfect square.

The symbolic logic is:

$$p \vee q.$$

- **Exercise 3**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

The sum of a and b is greater than zero or zero and smaller than the product of them.

- **Solution 3.1**

Let:

$p : \text{The sum of } a \text{ and } b \text{ is greater than zero } (a + b > 0).$

$q : \text{Zero is smaller than the product of them } (0 < ab).$

The symbolic logic is:

$$p \vee q.$$

- **Exercise 4**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

If P is a parabola, then the equation of P is quadratic.

- **Solution 4.1**

Let:

$p : P$ is a parabola.

$q : \text{The equation of } P \text{ is quadratic.}$

The symbolic logic is:

$$p \implies q.$$

- **Exercise 5**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:
 $x - 2$ is greater than zero and $x + 2$ is smaller than zero or $x - 2$ is smaller than zero and $x + 2$ is greater than zero.

◦ **Solution 5.1**

Let:

$p : x - 2$ is greater than zero ($x - 2 > 0$).

$q : x + 2$ is smaller than zero ($x + 2 < 0$).

$r : x - 2$ is smaller than zero ($x - 2 < 0$).

$s : x + 2$ is greater than zero ($x + 2 > 0$).

The symbolic logic is:

$$(p \wedge q) \vee (r \wedge s).$$

• **Exercise 6**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:
 $(x + 2)(x - 2)$ is smaller than zero.

◦ **Solution 6.1**

Let:

$p : (x + 2)(x - 2)$ is smaller than zero.

The symbolic logic is:

$$p.$$

• **Exercise 7**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:
It is not true that $x - 3$ is greater than zero or smaller than 1.

◦ **Solution 7.1**

Let:

$p : x - 3$ is greater than zero ($x - 3 > 0$).

$q : x - 3$ is smaller than 1 ($x - 3 < 1$).

The symbolic logic is:

$$\neg(p \vee q).$$

• **Exercise 8**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:
If f is not a constant, then f' is not zero.

◦ **Solution 8.1**

Let:

$p : f$ is a constant.

$q : f'$ is zero.

The symbolic logic is:

$$\neg p \implies \neg q.$$

• **Exercise 9**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

If $\sin x$ is 0 and x is greater than, or equal to 0 and smaller than or equal to 2π , then $x = 0$ or $x = \pi$.

◦ **Solution 9.1**

Let:

$p : \sin x$ is 0.

$q : x$ is greater than, or equal to 0 and smaller than or equal to 2π ($0 \leq x \leq 2\pi$).

$r : x = 0$.

$s : x = \pi$.

The symbolic logic is:

$$(p \wedge q) \implies (r \vee s).$$

• **Exercise 10**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

If $\cos x \cdot \sin x$ is 0 and x is greater than 0 and smaller than, or equal to π , then $x = \frac{\pi}{2}$.

◦ **Solution 10.1**

Let:

$p : \cos x \cdot \sin x$ is 0.

$q : x$ is greater than 0 and smaller than, or equal to π ($0 < x \leq \pi$).

$r : x = \frac{\pi}{2}$.

The symbolic logic is:

$$(p \wedge q) \implies r.$$

• **Exercise 11**

Let p stand for the proposition " $x^2 + 2x - 3$ is 0" and q for " $x^2 - x - 2$ is 0". Express the following as natural sentences in English:

$$\neg p.$$

$$p \wedge q.$$

$$p \vee q.$$

◦ **Solution 11.1**

$$x^2 + 2x - 3 \text{ is not } 0.$$

$$x^2 + 2x - 3 \text{ is } 0 \text{ and } x^2 - x - 2 \text{ is } 0.$$

$$x^2 + 2x - 3 \text{ is } 0 \text{ or } x^2 - x - 2 \text{ is } 0.$$

High school mathematical concepts with quantified definitions

• **Exercise 14**

Perfect square

An integer k is a perfect square if there exists an integer n such that $k = n^2$.

◦ **Solution 14.1**

The statement can be expressed as:

$$\exists n \in \mathbb{Z}, k = n^2.$$

• **Exercise 15**

Odd number

An integer m is odd if there exists an integer t such that $m = 2t + 1$.

- **Solution 15.1**

The statement can be expressed as:

$$(\exists t \in \mathbb{Z}) m = 2t + 1.$$

- **Exercise 16**

Multiple

An integer n is a multiple of the integer d if there exists an integer k such that $n = dk$.

- **Solution 16.1**

The statement can be expressed as:

$$(\exists k \in \mathbb{Z}), n = dk.$$

- **Exercise 17**

Rational number

A real number x is rational if there exist integers a, b with $b \neq 0$ such that $x = \frac{a}{b}$.

- **Solution 17.1**

The statement can be expressed as:

$$(\exists a \in \mathbb{Z})(\exists b \in \mathbb{Z}) b \neq 0, x = \frac{a}{b}.$$

- **Exercise 18**

Irrational number

A real number x is irrational if for all integers a, b with $b \neq 0$, we have $x \neq \frac{a}{b}$.

- **Solution 18.1**

The statement can be expressed as:

$$(\forall a, b \in \mathbb{Z}) b \neq 0 \implies x \neq \frac{a}{b}.$$

- **Exercise 19**

Root (zero) of a function

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ has a root if there exists a number $c \in \mathbb{R}$, such that $f(c) = 0$.

- **Solution 19.1**

The statement can be expressed as:

$$(\exists c \in \mathbb{R}) f(c) = 0.$$

- **Exercise 20**

Function bounded above

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ is bounded above if there exists a real number $M > 0$, such that for all $x \in \mathbb{R}$, $f(x) \leq M$.

- **Solution 20.1**

The statement can be expressed as:

$$(\exists M \in \mathbb{R}^+)(\forall x \in \mathbb{R}) f(x) \leq M.$$

- **Exercise 21**

- Function bounded below**

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ is bounded below if there exists a real number m such that for all $x \in \mathbb{R}$, $f(x) \geq m$.

- **Solution 21.1**

The statement can be expressed as:

$$(\exists m \in \mathbb{R})(\forall x \in \mathbb{R}) f(x) \geq m.$$

- **Exercise 22**

- Strictly increasing function**

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing if for all $x_1, x_2 \in \mathbb{R}$, $x_1 < x_2$, we have $f(x_1) < f(x_2)$.

- **Solution 22.1**

The statement can be expressed as:

$$(\forall x_1, x_2), x_1 < x_2 \implies f(x_1) < f(x_2).$$

- **Exercise 23**

- Equal functions**

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ equals a real function $g : \mathbb{R} \rightarrow \mathbb{R}$ if for every $x \in \mathbb{R}$, $f(x) = g(x)$.

- **Solution 23.1**

The statement can be expressed as:

$$(\forall x \in \mathbb{R}) f(x) = g(x).$$

- **Exercise 24**

- Different functions**

A real function $f : \mathbb{R} \rightarrow \mathbb{R}$ is different from a real function $g : \mathbb{R} \rightarrow \mathbb{R}$ if there exists $x \in \mathbb{R}$, $f(x) \neq g(x)$.

- **Solution 24.1**

The statement can be expressed as:

$$(\exists x \in \mathbb{R}) f(x) \neq g(x).$$

- **Exercise 25**

- Congruent triangles**

Two triangles are congruent if there exists a rigid transformation T mapping one triangle onto the other.

- **Solution 25.1**

The statement can be expressed as:

$$(\exists \text{rigid motion } T) T(\Delta_1) = \Delta_2.$$

- **Exercise 26**

- A very well-known Theorem of High-School**

Let $f(x) = ax^2 + bx + c$, $a, b, c \in \mathbb{R}$ be any quadric.

(a) If $b^2 - 4ac < 0$, then f doesn't have real roots.

(b) If $b^2 - 4ac = 0$, then f has exactly one real root.

(c) If $b^2 - 4ac > 0$, then f has exactly two real roots.

- **Solution 26.1**

For statement (a), it can be expressed as:

$$(\forall x \in \mathbb{R}) f(x) \neq 0.$$

◦ **Solution 26.2**

For statement (b), using the unique existential quantifier $\exists!$:

$$(\exists! x) f(x) = 0.$$

Alternatively, without using $\exists!$:

$$(\exists x)(\forall y) (f(x) = 0 \wedge (y \neq x \implies f(y) \neq 0)).$$

◦ **Solution 26.3**

For statement (c), it can be expressed as:

$$(\exists x)(\exists z)(\forall y) (x \neq z \wedge f(x) = 0 \wedge f(z) = 0 \wedge (y \neq x \wedge y \neq z \implies f(y) \neq 0)).$$